



Leica TCS SP5 II

The Only Broadband Confocal
Technical Documentation

Living up to Life

Leica
MICROSYSTEMS

Specifications

Microscopes	Upright	Leica DM6000 CS
		Leica DM6000 CFS
	Inverted	Leica DMI6000 CS
		Leica DMI6000 CS bottom port
Microscope anti-vibration table	Specification	For imaging
	Vibration insulation	Passive
Z-drive	SuperZ galvanometer stage	1500 μ m travel range/3 nm stepsize
	Motorfocus (stand)	Travel range depending on mechanics of microscope/15 nm step size
Continuous wave lasers	Laser type	For imaging
	VIS	WLL, average power 1.5 mW: 470–670 nm
		Diode, 40 mW: 442 nm
		Ar, 65 mW: 458, 476, 488, 496, 514 nm
		HeNe, 1 mW: 543 nm
		HeNe, 2 mW: 594 nm
		HeNe, 10 mW: 633 nm
		DPSS, 20 mW: 561 nm
		UV OPSSL, 80 mW: 355 nm
	UV	Diode, 50 mW: 405 nm
Pulsed lasers	Laser type	For imaging
	IR	TiSa (ps or fs) 1 W 690...1040 nm (various ranges)
	VIS	—
	UV	—
	UV	—
Excitation modulation	Modulation type	For imaging
	AOTF VIS	Up to 8 channels
	AOTF UV	Up to 3 channels
	EOM IR	Yes
	Pulsed laser driver	Optional

Specifications

Optics	Features	For imaging
	Number of laser ports	Up to 3 (UV - VIS - IR)
	Number of lasers	Up to 8
	Excitation – emission splitting	Acousto Optical Beam Splitter (AOBS®) or dichroic beam splitters
	Detection range	400...800 nm
	UV and IR imaging	Sequential or simultaneous
	Field upgradable	Yes (UV, IR)
	UV correction	Individual precise correction optics (up to 5 positions)
	Pinhole	Alignment stable single pinhole
	Pinhole diameter control	Motorized by software, automatic mode available
	Switchable beam expander, optional	for DM6000/DML6000: available for DM6000 CFS: not available
	Notch filters, optional	458 / 514 nm 488 / 561 / 633 nm
Scanner	Scanner design	For imaging
	Scanning concept	Optically correct scanning at low inertia
	Switch conventional – resonant scanner	Conventional and resonant scanner in one system (optional)
	Conventional scanner	For imaging (PMT and APD)
	Maximal line frequency	2800 Hz
	Minimal line frequency	1 Hz
	Scan speed granulation	1400
	Maximal frame rate 512 x 512	5 Hz
	Maximal frame rate 512 x 16	50 Hz
	Beam park	Yes
	Maximal frame resolution	8192 x 8192 pixel
	Scan zoom	1.0 ... 64 x
	Panning	Yes
	Field rotation	200° optical
	Field diameter	22 mm

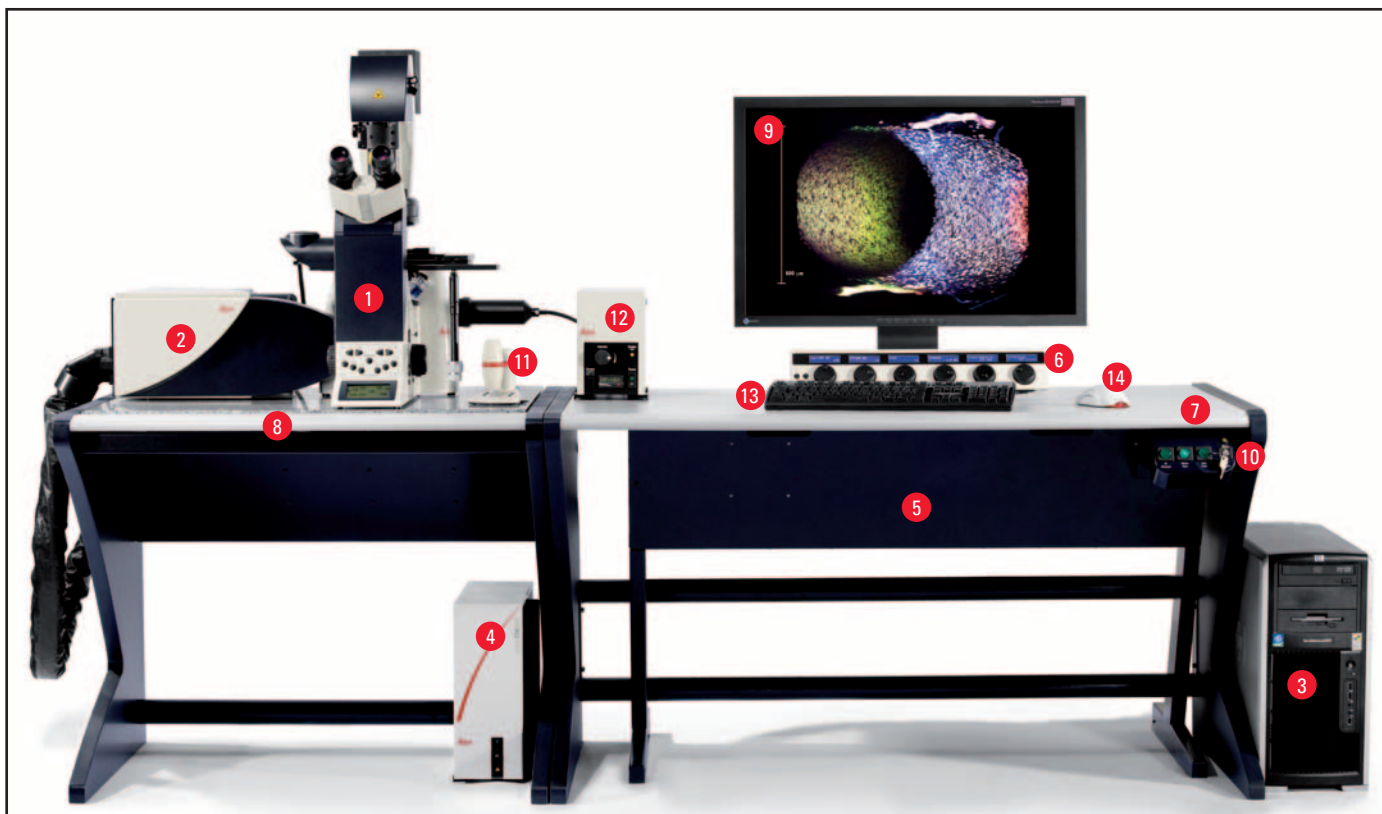
Specifications

Scanner	Resonant scanner	For imaging
	Maximal line frequency	16000 Hz
	Minimal line frequency	8000 Hz
	Scan speed granulation	1
	Maximal frame rate 512 x 512	28 Hz
	Maximal frame rate 512 x 16	290 Hz
	Beam park	No
	Maximal frame resolution	1024 x 1024 pixel
	Scan zoom	1.7 ... 64 x
	Panning	Yes
	Field rotation	200° optical
	Field diameter	15 mm
Scan modes	Scan options	For imaging
	xt	Yes
	xy	Yes
	xyt	Yes
	xyλ	Yes
	xz	Yes
	xzλ	Yes
	xyz	Yes
	xyzλ	Yes
	xyt	Yes
	xzt	Yes
	xyzt	Yes
	xytz	Yes

Specifications		
Internal confocal detection	Detection features	Up to 5 PMT for confocal imaging
	Emission separation	Highly sensitive prism spectral detector
	Maximum number of confocal channels	5
	Tunability of emission bands	Yes
	Spectral detection range	400 – 800 nm
	Tuning steps of emission bands	1 nm
	Minimal detection range	5 nm
	Sensors	High sensitivity low noise PMT: R 9624
	Digitization	12 or 18 bit per channel
	Max. grey resolution	16 bit imaging
	Read out frequency	40 MHz
External confocal detection	Detection features	2 APDs for confocal imaging
	Emission separation	User-exchangeable beam splitting filter cubes
	Confocal channels	2
	Sensors	APDs from PE (SPCM-AQRH series) or MPD (PDM series)
	Quantum efficiency	PE APD: wavelength dependent, typ. 65% @ 670 nm MPD APD: wavelength dependent, typ. 45% @ 550 nm
	Dark counts	PE APD: < 250 cps MPD APD: < 250 cps
	Jitter FWHM	Not relevant
	Dead time	Not relevant
Non-confocal detection	Detection types	For Imaging
	Transmitted light detector	Optional, allowing BF, DIC, Ph etc.
	Non descanned transmitted light channels	Up to 4 channels (MP)
	Non descanned reflected light channels	Up to 4 channels (MP)
Electronics	Devices	For imaging
	Scanner control	Digitally at high performance (FPGA, field programmable gate arrays)
	Trigger in/out functions	Yes
	Auxiliary data input channels	Up to 2
	Max channels in parallel	12
	Computer	High performance PC workstation
	Monitors	2 x 19" monitors or 1 x 30" monitor
	Integration of third party software	—
	—	Programmable control panel with LCD function & value display

Specifications

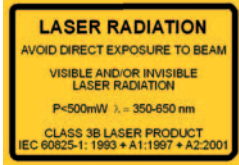
Extensions	Devices	For imaging
	Fast ROI-spectrometer	Optional
	Auxiliary emission port	Optional
	Environment accessories	Various options
Software (LAS AF)	General	Intuitive and guiding user interface
	Context sensitive online help system	Included
	Multi-dimensional data acquisition	Included
	Region of interest (ROI) scan	Included
	Excitation line/frame sequential scan	Included
	Emission spectrum recording	Included
	Quantification tools	Included
	Multi-color restoration, spectral unmixing	Included
	General time lapse experiment control tile scanning (mosaic scan)	Included
Software options (LAS AF)	Dedicated application wizards	For imaging
	Live Data Mode	Interactive data recording also allowing job sequencing and online evaluation
	Advanced Mark & Find	Combines Mark & Find with sophisticated 3D recordings, Live Data Mode etc.
	3D visualization	Maximum and other projections, simulated fluorescence process, rotation animations, stereo pairs, red-green anaglyphs, height color coded extended depth of focus images etc.
	Colocalization	Histogram based colocalization and area measurements
	Deconvolution	Deconvolution option for widefield and confocal images
	MicroLab	FRAP wizard, FRAPxt wizard, FLIP wizard, FRET SE wizard, FRET AB wizard etc.
	SMD FCS wizard	—
	SMD FLIM wizard	—
	Electrophysiology	Interactive data recording also allowing correlation of optical and electrical data



- ① Research Microscope
- ② Scanhead
- ③ Workstation
- ④ Microscope Control Unit
- ⑤ Laser Supply and Power Supply
- ⑥ Control Panel
- ⑦ Computer Table
- ⑧ Anti-vibration Table
- ⑨ Monitor
- ⑩ Supply Control
- ⑪ Smart Move
- ⑫ EL6000 Fluorescence Illumination
- ⑬ Keyboard
- ⑭ Computer Mouse



visible and ultraviolet
radiation:



infrared radiation:



Installation Requirements

Weight base system: VIS: max. 320 kg
UV: max. 428 kg
IR: Optical bench 900 x 1500 mm + ca. 280 kg
IR Laser System + ca. 100 kg

Heat load max.: VIS: 3,2 kW
UV: 0,5 kW
IR: 1.5 kW

Separate cooling: UV laser, air-cooled heat exchanger
IR laser, air-cooled heat exchanger (chiller)

Electric apply: VIS lasers: 100 ... 240 V AC $\pm$ 10 %
2 x 1600 VA, 50/60 Hz (Power input 1+2)

UV laser: 100 ... 240 V AC $\pm$ 10 %
750 VA, 50/60 Hz

IR laser: 100 ... 240 V AC $\pm$ 10 %
15 ... 10 A, 50/60 Hz

Chiller for IR laser: 110 V/230 V AC $\pm$ 10 %
10 A/6 A, 50/60 Hz

Environment: Room temperature: + 18 ... + 25 °C
Avoid proximity to air conditioning equipment
Protect from dust
Room darkening recommended

